



Network ACL vs Security Group

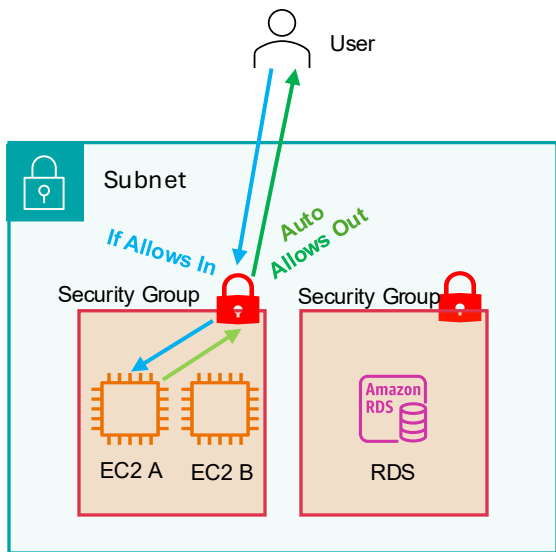
- Alon Shrestha



Security Group



- Instance-level firewall
- Controls inbound & outbound traffic
- **Stateful:** if request allowed, responses auto-allowed



Security Group



Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info
HTTP ▼	TCP	80	Custom ▼
			Q 10.0.1.12/32
			10.0.1.12/32 X

[Add rule](#)

Outbound rules [Info](#)

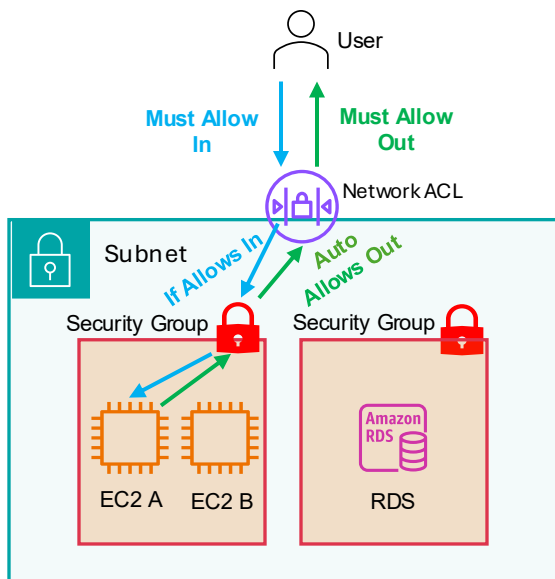
This security group has no outbound rules.

[Add rule](#)

Network ACL



- Subnet-level firewall
- Controls inbound & outbound traffic
- **Stateless:** requests & responses must be allowed separately



Network ACL



Inbound rules (2)

Q Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	HTTP (80)	TCP (6)	80	10.0.1.12/32	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Outbound rules (2)

Q Filter outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	HTTP (80)	TCP (6)	80	10.0.1.12/32	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



User



Send
Request

Receive
Response



Internet Gateway



VPC

Routes

Routes

172.16.0.0
172.16.1.0
172.16.2.0

Route Table

Must Allow
In

Must Allow
Out



NetworkACL



Subnet

If Allows In

Auto
Allows Out

If No Allows In

Security group A

Response



EC2-A



EC2-B

Security group B

EC2-C



RDS



User



Send Request

Request Rejected



Internet Gateway



VPC

Routes

172.16.0.0
172.16.1.0
172.16.2.0

Route Table

Must Allow In

Not Allow Out



NetworkACL



Subnet

If Allows In

Auto Allows Out

If No Allows In

Security group A

Response



EC2-A



EC2-B

Security group B



EC2-C



RDS

Thank you!